Online PES Decoder Learning in Neuromorphic Residual Control of UAV Quaternion Tracking

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***Abstract*— Online neuromorphic residual control can augment a classical attitude controller under abrupt missions or unforeseen disturbances without changing its feedback gains or actuator interface. We present six continuous coupled membrane-gate (CMG) units and eighteen decoder weights adapted by prescribed error sensitivity (PES), adding a bounded correction to the saturated quaternion proportional-derivative (PD) torque request before rotor-force allocation. Across ten matched 45-second MuJoCo cases on the Skydio X2 and a Crazyflie 2 derived model, mean quaternion-angle error decreases by 17.3–60.6%, including approximately 53–57% reductions on the Full3 and Hard45 multi-axis tracking missions. Direct and Nengo implementations agree closely. Benchmarks against LIF, ALIF, Izhikevich, and LIFRate neuron models expose an effort-accuracy tradeoff: CMG–PES attains better tracking with substantially lower realized residual effort. Under synthetic measurement noise, the error reductions persist, and learned corrections are reused across repeated disturbances. Overall, PD provides a nominal, interpretable reference, while the online residual adds bounded, revocable adaptation that improves robustness to disturbances and noisy measurements.**

I. Introduction

Learning-based augmentation can retain classical feedback while adapting a correction that is difficult to specify *a priori*. We study a bounded torque residual for quaternion attitude tracking of an unmanned aerial vehicle (UAV) with individual rotor-force saturation. The question is whether a small stateful basis improves a fixed PD controller, and whether any improvement comes from its neuron dynamics, decoder learning, or correction magnitude. Bounded authority makes the learning layer switchable; it does not by itself ensure stability or safety.

Prior work includes residual robot learning [1], geometric $\mathcal{L}_1$ quadrotor augmentation with performance guarantees [2], and proportional–integral–derivative (PID) attitude control augmented with a proximal policy optimization (PPO) residual [3]. Brain-inspired robust UAS controllers likewise pair model-based feedback with learned terms, from limbic-system designs [4], [5] to neural weight reset under impulsive disturbances [6], and perturbation estimation targets the same disturbances on small quadrotors [7]. Neuromorphic control offers a complementary implementation approach: Nengo supports the Neural Engineering Framework (NEF) [8] and online decoder learning [9], most often through the prescribed error sensitivity (PES) rule [10], an established local, online error-driven update [11]. PES-based adaptive control has been demonstrated on manipulators [12], [13], and in self-tuning spiking PID controllers for free-flying UAS trajectory tracking [14], [15]. Building on that line, our focus is a continuous membrane-gate basis for a bounded quaternion-torque residual, with a fixed classical controller and adaptation confined to its decoder.

We augment a quaternion PD baseline with a compact neuromorphic residual architecture based on coupled conductance-based membrane-gate (CMG) units and PES decoder learning. The paper's contributions are:

1) a code-aligned CMG–PES architecture with paired currents, explicit encoding, state-update, and decoder/readout equations, and a fully specified actuator interface (Fig. 1);
2) proofs of internal-state, command-deviation, and weight boundedness, with exact recovery of the PD baseline at zero residual authority, and an allocation-aware local input-to-state certificate whose constants are computed for both vehicles;
3) a controlled ablation that separates conductance modulation, decoder learning, and output authority, and identifies which mechanism is primarily responsible for the tracking-error reduction; and
4) robustness evidence that the reduction survives realistic measurement noise, together with a hold-equilibrium study in which the learned correction is reused across repeated disturbance episodes rather than relearned.

The CMG unit couples a leaky membrane to a gating state that modulates its effective conductance, linking LIF-like membrane integration with Hodgkin–Huxley inspired gating dynamics [16]. Although related operating regimes can support spike-like responses, the present controller deliberately exploits the continuous, non-spiking membrane-gate trajectory, which provides a compact set of stateful temporal features for the residual. The present implementation is digital; the same temporal-feature mechanism holds regardless of computing platform, and no neuromorphic-hardware energy advantage is claimed.

Section II defines the problem and actuator interface; Section III presents the CMG–PES architecture, its boundedness properties, and a local stability certificate; Section IV reports the simulation study, ablations, benchmarks, and noise and disturbance experiments; and Section V discusses limitations and future work. Code, model files, and run archives will be available at `https://github.com/[to-be-released]`.

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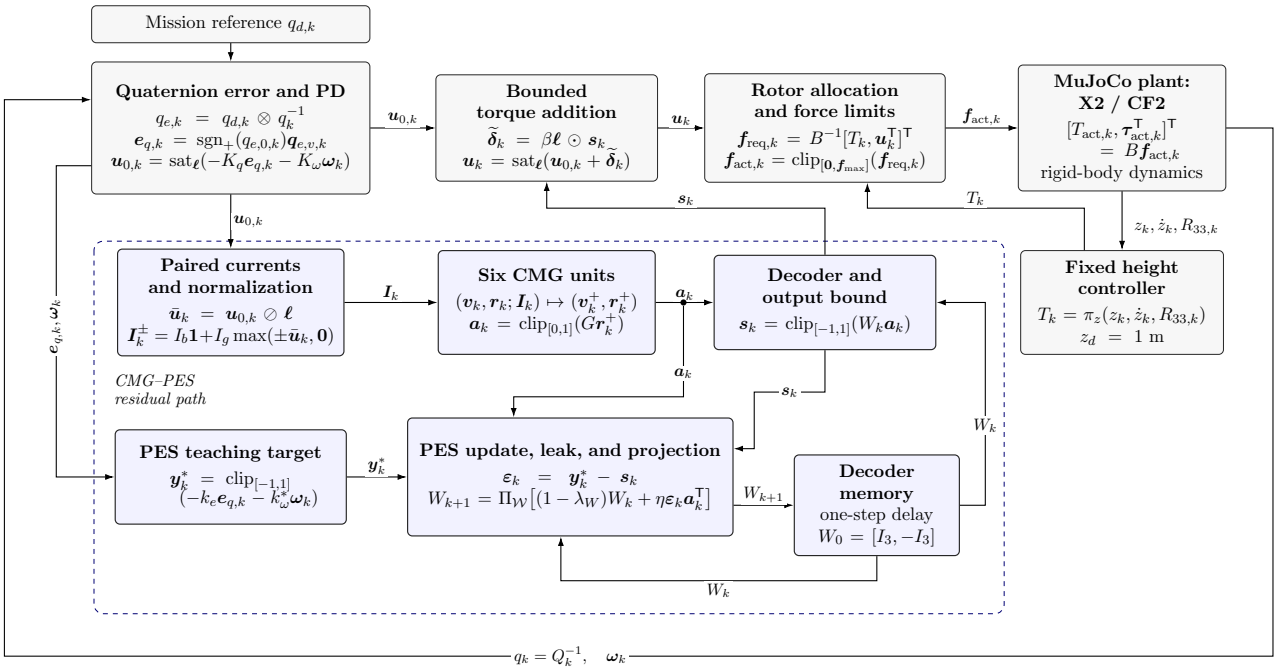


Fig. 1. Online PES decoder learning CMG-PES Neuromorphic Residual Control architecture



Fig. 2. Vehicles and simulation models. Left: CF2 hardware and MuJoCo model. Centre: X2 hardware and MuJoCo model. Right: both geometries at relative scale. The axis triads are body-frame axes: they are rigidly attached to the vehicle and therefore rotate with it relative to the inertial (world) frame (cf. Q: body $\to$ inertial in Sec. II).

II. Attitude Baseline and Actuator Interface

The objective is to reduce shortest-path attitude error for a prescribed $q_d(t)$ while respecting rotor-force limits. Baseline and augmented controllers share the state measurements, height loop, and allocator; Section III modifies only the requested attitude torque.

A. Quaternion Convention and Error

Let $Q \in \mathbb{S}^3$ denote the scalar-first Hamilton quaternion representing attitude (body to inertial), and let $\boldsymbol{\omega}$ be the body angular velocity. For a rigid body with inertia $\boldsymbol{J} \succ 0$ about the center of mass, expressed in body axes (cf. Fig. 2), the free-flight rotational dynamics are [17]

$$\dot{Q} = \tfrac{1}{2} Q \otimes [0, \boldsymbol{\omega}^\mathsf{T}]^\mathsf{T}, \tag{1}$$

$$\boldsymbol{J}\dot{\boldsymbol{\omega}} = \boldsymbol{\tau}_{\mathrm{act}} + \boldsymbol{\tau}_{\mathrm{p}} + \boldsymbol{\tau}_{\mathrm{d}} - \boldsymbol{\omega} \times (\boldsymbol{J}\boldsymbol{\omega}). \tag{2}$$

Here $\boldsymbol{\tau}_{\mathrm{p}}$ denotes modeled passive torque and $\boldsymbol{\tau}_{\mathrm{d}}$ any applied disturbance; the nominal ideal rigid-body model sets both to zero. Contact, if present, contributes an additional external wrench.

We work with the corresponding passive quaternion $q = Q^{-1}$ [18], which satisfies

$$\dot{q} = -\tfrac{1}{2}[0, \boldsymbol{\omega}^\mathsf{T}]^\mathsf{T} \otimes q. \tag{3}$$

The desired attitude q_d uses the same convention. Define the attitude error

$$q_e = q_d \otimes q^{-1} = [q_{e,0}, \boldsymbol{q}_{e,v}^\mathsf{T}]^\mathsf{T}, \tag{4}$$

$$\boldsymbol{e}_q = \mathrm{sgn}_+(q_{e,0}) \boldsymbol{q}_{e,v}, \qquad \mathrm{sgn}_+(0) = 1, \tag{5}$$

$$\theta_q = 2 \operatorname{atan2}(\|\boldsymbol{q}_{e,v}\|_2, |q_{e,0}|) \in [0, \pi]. \tag{6}$$

The reported error θ_q is the shortest rotation angle, invariant under $q_e \mapsto -q_e$. Sign selection introduces a discontinuity at $q_{e,0} = 0$; no continuous feedback globally stabilizes rigid-body attitude [19], [20], robust global tracking requires hybrid [21] or geometric-algebra [22], [23] formulations, and the analysis here is local.

For a desired-frame rate $\boldsymbol{\omega}_d$ satisfying the corresponding

passive kinematics, differentiation gives

$$\dot{q}_e = \tfrac{1}{2} q_e \otimes [0, \boldsymbol{\omega}_e^{\mathsf{T}}]^{\mathsf{T}}, \tag{7}$$

$$\boldsymbol{\omega}_e = \boldsymbol{\omega} - R(q_e)^{\mathsf{T}} \boldsymbol{\omega}_d, \qquad \dot{q}_{e,0} = -\tfrac{1}{2} \boldsymbol{q}_{e,v}^{\mathsf{T}} \boldsymbol{\omega}_e. \tag{8}$$

These are kinematic identities (no convergence implied). Reference Euler triples parameterize q_d in this passive convention.

B. Physical Inertia and Implemented PD Gains

When principal moments $\boldsymbol{j}_{\mathrm{p}}$ of $\boldsymbol{J}$ are specified in a rotated principal-axis frame, the physical body-frame tensor is

$$\boldsymbol{J} = R_{BI} \operatorname{diag}(\boldsymbol{j}_{\mathrm{p}}) R_{BI}^{\mathsf{T}}, \tag{9}$$

for the rotation from principal axes to body axes R_{BI}. The implementation sets axiswise gains using the principal moments in their reported ordering, $\boldsymbol{j}_{\mathrm{tune}}$, not necessarily $\operatorname{diag}(\boldsymbol{J})$ (Table I).

The fixed quaternion PD baseline is

$$\boldsymbol{k}_q = 2\boldsymbol{j}_{\mathrm{tune}} \odot \boldsymbol{\omega}_n^2, \qquad \boldsymbol{k}_\omega = 2\zeta \boldsymbol{j}_{\mathrm{tune}} \odot \boldsymbol{\omega}_n, \tag{10}$$

$$\boldsymbol{u}_0 = \mathrm{sat}_{\boldsymbol{\ell}}(-\boldsymbol{k}_q \odot \boldsymbol{e}_q - \boldsymbol{k}_\omega \odot \boldsymbol{\omega}), \tag{11}$$

where $\odot$ denotes componentwise multiplication, $\boldsymbol{\omega}_n = (4, 4, 3.5)^{\mathsf{T}}$ rad/s, $\zeta = 0.95$, and $\mathrm{sat}_{\boldsymbol{\ell}}$ clips component i to $[-\ell_i, \ell_i]$. The requested-torque box is $\mathcal{U} = \prod_{i=1}^{3} [-\ell_i, \ell_i]$; $\mathrm{clip}_{[a,b]}$ also acts componentwise, and $\oslash$ denotes componentwise division. We impose componentwise torque limits $\boldsymbol{\ell} \in \mathbb{R}^3_{>0}$ through $\mathrm{sat}_{\boldsymbol{\ell}}(\cdot)$. In our implementation, $\boldsymbol{\ell}$ is chosen proportional to the diagonal gain-scheduling vector $\boldsymbol{j}_{\mathrm{tune}}$ via $\ell_i = \max(j_{\mathrm{tune},i} \nu_i, 10^{-6}\ \mathrm{N\,m})$, with acceleration-scale vector $\boldsymbol{\nu} = (12, 12, 10)^{\mathsf{T}}$ rad/s^2. These controller-side bounds are not the rotor-feasible torque set. The law is quaternion feedback [24]; $\boldsymbol{\omega}_n$ is a tuning parameter rather than a claimed set of coupled closed-loop natural frequencies. Desired-rate feedforward and gyroscopic cancellation are omitted. The resulting gains are vehicle-specific through $\boldsymbol{j}_{\mathrm{tune}}$, even though $\boldsymbol{\omega}_n$ and ζ are shared; the same fixed gains are then retained for PD and its CMG augmentation. The local result in Section III-D uses these anisotropic gains; it does not replace them with a scalar-gain controller.

C. Requested and Realized Wrench

For rotor positions $\boldsymbol{p}_i$ in body coordinates, parallel $+z$ thrust axes, and signed reaction coefficients κ_i, define

$$B = \begin{bmatrix} 1 & 1 & 1 & 1 \\ p_{1,y} & p_{2,y} & p_{3,y} & p_{4,y} \\ -p_{1,x} & -p_{2,x} & -p_{3,x} & -p_{4,x} \\ \kappa_1 & \kappa_2 & \kappa_3 & \kappa_4 \end{bmatrix}. \tag{12}$$

The signs follow $\boldsymbol{p}_i \times [0, 0, f_i]^{\mathsf{T}}$ and the actuator's local reaction torque $\kappa_i f_i$ [25]. Here B maps rotor thrusts to total thrust and body torque; κ_i has units of meters, not angular velocity. For four rotors B^{-1} equals the Moore–Penrose inverse $B^{\dagger}$ that other rotor counts use. Given a commanded collective thrust T and body-torque command $\boldsymbol{u}$, the corresponding inverse-allocation and clipping model is

$$\begin{aligned} \boldsymbol{f}_{\mathrm{req}} &= B^{-1}[T, \boldsymbol{u}^{\mathsf{T}}]^{\mathsf{T}}, \\ \boldsymbol{f}_{\mathrm{act}} &= \mathrm{clip}_{[\boldsymbol{0}, \boldsymbol{f}_{\max}]}(\boldsymbol{f}_{\mathrm{req}}), \\ [T_{\mathrm{act}}, \boldsymbol{\tau}_{\mathrm{act}}^{\mathsf{T}}]^{\mathsf{T}} &= B \boldsymbol{f}_{\mathrm{act}}. \end{aligned} \tag{13}$$

Here $\boldsymbol{u} = \boldsymbol{u}_0$ for PD; the CMG path in Section III modifies that requested torque before the same allocation and clipping. Each f_i is a commanded force in newtons, not a rotor-speed command. Exact wrench realization requires $B^{-1}[T, \boldsymbol{u}^{\mathsf{T}}]^{\mathsf{T}} \in [\boldsymbol{0}, \boldsymbol{f}_{\max}]$. A componentwise torque box is not generally contained in this collective-dependent feasible set. Clipped inverse allocation is used, not an optimization-based allocator [26].

Collective thrust remains classical, with $z_d = 1$ m:

$$T = \mathrm{clip}_{[0, 2.5Mg]} \left(\frac{M[g + 6(z_d - z) - 4\dot{z}]}{\max(R(Q)_{33}, 0.35)} \right). \tag{14}$$

Here g is gravity; the height loop reads altitude and vertical velocity, and the denominator compensates tilt. The guard 0.35 avoids near-singular compensation as $R(Q)_{33} \to 0$, limiting it to about $\arccos(0.35) \approx 69.5°$; the upper clip does not supersede individual rotor limits. The CMG path has no direct collective output, but can affect height through attitude, tilt compensation, and allocation saturation. The resulting shared plant interface isolates changes in requested attitude torque while preserving the same physical simulation for both controllers.

III. CMG State Dynamics and Online Decoder Adaptation

This section specifies the residual architecture in state-space form and the online decoder update, then states the boundedness properties that follow from the resulting saturations and projections.

A. Encoding, States, and Numerical Update

For m torque channels and $2m$ units, normalize the clipped baseline and form positive/negative current pairs:

$$\begin{aligned} \bar{\boldsymbol{u}} &= \boldsymbol{u}_0 \oslash \boldsymbol{\ell}, \\ \boldsymbol{I}^{\pm} &= I_b \boldsymbol{1} + I_g \max(\pm \bar{\boldsymbol{u}}, \boldsymbol{0}), \\ \boldsymbol{I} &= [(\boldsymbol{I}^{+})^{\mathsf{T}}, (\boldsymbol{I}^{-})^{\mathsf{T}}]^{\mathsf{T}}. \end{aligned} \tag{15}$$

Each unit has a dimensionless membrane state v_j and gate r_j:

$$\dot{r}_j = [\sigma(\alpha_h(v_j - v_{\mathrm{th}} + \gamma r_j)) - r_j]/\tau_r, \tag{16}$$

$$\dot{v}_j = -(v_j - v_{\mathrm{rest}})/\tau + I_j/\tau - c_j(v_j - v_{\mathrm{reset}}), \tag{17}$$

$$c_j = (g_0 + \chi g_1 |I_j|) r_j^2, \qquad \chi \in \{0, 1\}. \tag{18}$$

The logistic gate target depends on v_j, while r_j shifts that target and modulates membrane conductance (Fig. 3). Thus, both states participate even though the decoder reads the gate-derived activity. The model has no threshold-triggered reset or spike event; v_{reset} is a reversal level in a continuous conductance term. The gains g_0, g_1 have the units needed to make c_j a rate.

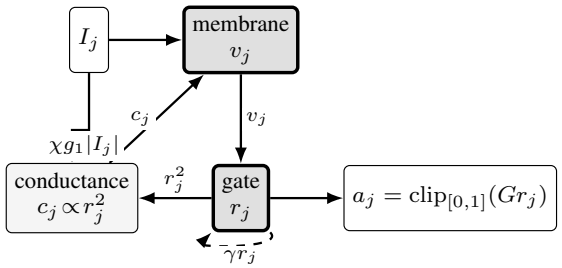


Fig. 3. Coupling topology of one CMG unit, cf. (18). The membrane sets the gate target; the gate acts back twice, shifting its own threshold through γ and shunting the membrane through $c_j \propto r_j^2$ ($\chi = 1$ makes that shunt input dependent). The dynamics are continuous: no spike, no reset, and only a_j leaves the unit.

The actual integration is sequential, not exponential Euler for both states. From the old states compute

$$b_{j,k} = \sigma\Big(\mathrm{clip}_{[-60,60]}[\alpha_h(v_{j,k} - v_{\mathrm{th}} + \gamma r_{j,k})]\Big), \tag{19}$$

$$r^+_{j,k} = \mathrm{clip}_{[0,1]}[r_{j,k} + (h/\tau_r)(b_{j,k} - r_{j,k})]. \tag{20}$$

Then evaluate $c_{j,k} = (g_0 + \chi g_1|I_{j,k}|)(r^+_{j,k})^2$, $\lambda_{j,k} = \tau^{-1} + c_{j,k}$, and $d_{j,k} = (v_{\mathrm{rest}} + I_{j,k})/\tau + c_{j,k}v_{\mathrm{reset}}$. The frozen-coefficient membrane step is

$$v^+_{j,k} = e^{-\lambda_{j,k}h}v_{j,k} + (1 - e^{-\lambda_{j,k}h})d_{j,k}/\lambda_{j,k}. \tag{21}$$

This exactly solves the scalar membrane equation with those frozen coefficients; it does not establish unconditional stability of the coupled nonlinear closed loop. Assign $(v_{j,k+1}, r_{j,k+1}) = (v^+_{j,k}, r^+_{j,k})$. The exponential-Euler step $r^+_{j,k} = e^{-h/\tau_r}r_{j,k} + (1 - e^{-h/\tau_r})b_{j,k}$, exact for a frozen target, is convex for every $h > 0$ without a clip [27]; it changes only the weight on $r_{j,k}$ (0.905 vs. 0.9 at $h/\tau_r = 0.1$), and all results use (20).

B. Readout, PES, and Additive Residual

The within-step activity, initial decoder, and clipped readout are

$$\begin{aligned} \boldsymbol{a}_k &= \mathrm{clip}_{[0,1]}(G\boldsymbol{r}^+_k), \qquad W_0 = [I_m, -I_m], \\ \boldsymbol{s}_k &= \mathrm{clip}_{[-1,1]}(W_k\boldsymbol{a}_k). \end{aligned} \tag{22}$$

Pair subtraction retains directional information from a nonnegative activity basis; state memory does not imply instantaneous sign agreement after a command reversal. With $G = 10^5$ the activity saturates once $r_j \geq 10^{-5}$, so G is a primary design hyperparameter for how much of r_j's dynamic range reaches the decoder. The teaching target and local update are

$$\boldsymbol{y}^*_k = \mathrm{clip}_{[-1,1]}(-k_e\boldsymbol{e}_{q,k} - k^*_\omega\boldsymbol{\omega}_k), \tag{23}$$

$$\boldsymbol{\varepsilon}_k = \boldsymbol{y}^*_k - \boldsymbol{s}_k, \tag{24}$$

$$W_{k+1} = \Pi_{\mathcal{W}}[W_k + \eta\boldsymbol{\varepsilon}_k\boldsymbol{a}_k^{\mathsf{T}} - \lambda_W W_k]. \tag{25}$$

Here $\mathcal{W} = [-W_{\max}, W_{\max}]^{m\times 2m}$. The coefficients η and λ_W are per-step (so changing h changes the effective learning dynamics). PES is a local error-driven update toward the prescribed target $\boldsymbol{y}^*_k$; it does not differentiate through the plant or claim to minimize a closed-loop cost. Since $\boldsymbol{s}_k$ is normalized by the authority $\beta\boldsymbol{\ell}$, anisotropic scaling enters only through $\boldsymbol{\ell}$ and one pair (k_e, k^*_ω) serves all axes.

Using W_k before the update, the requested command is

$$\boldsymbol{u}_k = \mathrm{sat}_{\boldsymbol{\ell}}(\boldsymbol{u}_{0,k} + \beta\boldsymbol{\ell}\odot\boldsymbol{s}_k), \quad \boldsymbol{\delta}_k = \boldsymbol{u}_k - \boldsymbol{u}_{0,k}, \quad 0 \leq \beta \leq 1. \tag{26}$$

The canonical UAV case has $m = 3$, six CMG units, and eighteen decoder weights. Intrinsic conductance modulation (χ), synaptic/decoder adaptation (W_k), and output authority ($\beta\boldsymbol{\ell}$) are distinct.

C. Structural Properties and Their Scope

We first state the boundedness properties that follow directly from the architecture: internal-state bounds, a uniform command-deviation bound set by residual authority, and projection-based bounds on decoder weights.

Proposition 1 (Bounded internal states)**.** *With the nonnegative conductance parameters in Table I and $r_{j,0} \in [0,1]$, the gate remains in $[0,1]$ for every $h > 0$, and the gate clip in* (20) *is inactive when $0 < h \leq \tau_r$. The membrane iterate remains in the convex hull of its initial value, v_{reset}, and the interval $[v_{\mathrm{rest}} + I_b, v_{\mathrm{rest}} + I_b + I_g]$.*

Proof. For $0 < h \leq \tau_r$ the unclipped gate step is a convex combination of $r_{j,k}$ and $b_{j,k} \in [0,1]$, so the clip is inactive; for $h > \tau_r$ the clip enforces the bound. Since $I_j \in [I_b, I_b + I_g]$ and $c_j \geq 0$, d_j/λ_j is a convex combination of $v_{\mathrm{rest}} + I_j$ and v_{reset}. Equation (21) is a convex combination of $v_{j,k}$ and d_j/λ_j. Induction proves both statements. □

Proposition 2 (Bounded authority and baseline recovery)**.** *For $\boldsymbol{u}_{0,k} \in \mathcal{U}$ and any finite W_k, $\boldsymbol{u}_k \in \mathcal{U}$ and $|\delta_{k,i}| \leq \beta\ell_i$. If $\beta = 0$, the requested baseline command is recovered at every step. If $W_k = 0$ at a particular step, it is recovered at that step, but learning need not keep $W_{k+1} = 0$.*

Proof. The outer clip enforces membership in $\mathcal{U}$. Its componentwise nonexpansiveness and $\mathrm{sat}_{\ell_i}(u_{0,k,i}) = u_{0,k,i}$ give $|\delta_{k,i}| \leq |\beta\ell_i s_{k,i}| \leq \beta\ell_i$. The recovery statements follow by substitution. □

With $\beta = 0$ the same allocator and state give the same actuator output. The proposition establishes neither state safety, rotor headroom, stability, nor a bound on tracking improvement; a bounded adversarial correction can still degrade or destabilize a nonlinear plant.

Proposition 3 (Decoder bound and restricted contraction)**.** *If $W_0 \in \mathcal{W}$, then* (25) *keeps $W_k \in \mathcal{W}$. If, additionally, the target and a nonzero activity $\boldsymbol{a}$ are fixed, leak is zero, and readout clipping and projection remain inactive, then*

$$\boldsymbol{\varepsilon}_{k+1} = (1 - \eta\|\boldsymbol{a}\|_2^2)\boldsymbol{\varepsilon}_k. \tag{27}$$

For a nonzero initial error, its norm contracts to zero when $0 < \eta\|\boldsymbol{a}\|_2^2 < 2$.

Proof. Projection gives the first result. Under the additional conditions, $(W_{k+1} - W_k)\boldsymbol{a} = \eta\boldsymbol{\varepsilon}_k\|\boldsymbol{a}\|_2^2$; subtracting this from the fixed target error gives (27). The scalar multiplier has modulus less than one under the stated condition. □

Beyond those conditions, the readout clip is a nonexpansive projection onto a convex set containing $\boldsymbol{y}_k^*$ [28], so $\|\operatorname{clip}_{[-1,1]}(\boldsymbol{z}) - \boldsymbol{y}_k^*\| \le \|\boldsymbol{z} - \boldsymbol{y}_k^*\|$: saturation never enlarges the PES error. This is the constant-feature PES result [11], not a closed-loop convergence theorem. Since $\|\boldsymbol{a}\|^2 \le 2m$, $0 < \eta < 1/m$ is sufficient for that restricted model with nonzero activity; changing targets, loss of excitation, clipping, and leakage invalidate a direct application to the experiment. No unique identification of all decoder weights follows from one fixed regressor.

D. Allocation-Aware Local Stability

Let $E = [\mathbf{0}_{3\times1}, I_3]^\mathsf{T}$, $H = E^\mathsf{T}$, and let $\mathcal{A}_T(\boldsymbol{u}) = HB\operatorname{clip}_{[0,\boldsymbol{f}_{\max}]}(B^{-1}[T, \boldsymbol{u}^\mathsf{T}]^\mathsf{T})$ be the realized torque at collective T. For two commands at the *same state and collective*, scalar clipping has secant slopes $d_i \in [0,1]$. Consequently,

$$\mathcal{A}_T(\boldsymbol{u}) - \mathcal{A}_T(\boldsymbol{u}_0) = HBDB^{-1}E\boldsymbol{\delta}, \tag{28}$$

$$\|\mathcal{A}_T(\boldsymbol{u}) - \mathcal{A}_T(\boldsymbol{u}_0)\|_2 \le L_A\beta\|\boldsymbol{\ell}\|_2, \tag{29}$$

where $D = \operatorname{diag}(d_i)$ and $L_A = \max_{D\in\mathcal{D}}\|HBDB^{-1}E\|_2$. $\mathcal{D}$ is the set of sixteen diagonal matrices with entries in $\{0,1\}$. The vertex maximum bounds all intermediate D by convexity. Numerically $L_A = 4.53$ (X2) and 3.33 (CF2), inflated because small $|\kappa_i|$ turn clipped yaw requests into roll/pitch torque. If all of $\mathcal{U}$ is rotor-feasible at the operating collective (CF2 at hover, not X2), $D = I$ and 1 replaces L_A; a yaw-weighted norm helps only if Theorem 1 is restated in those coordinates. This holds even with rotor clipping, but is not a bound between two different closed-loop trajectories. It also does not bound the height error.

For a constant reference and a smooth shortest-path branch, let $\boldsymbol{x} = [\boldsymbol{e}_q^\mathsf{T}, \boldsymbol{\omega}^\mathsf{T}]^\mathsf{T}$ and $K_q = \operatorname{diag}(\boldsymbol{k}_q)$, $K_\omega = \operatorname{diag}(\boldsymbol{k}_\omega)$. Write the continuous-time attitude subsystem as

$$\dot{\boldsymbol{x}} = A\boldsymbol{x} + \boldsymbol{g}(\boldsymbol{x}) + F\boldsymbol{d},$$
$$A = \begin{bmatrix} 0 & \frac{1}{2}I_3 \\ -\boldsymbol{J}^{-1}K_q & -\boldsymbol{J}^{-1}K_\omega \end{bmatrix}, \qquad F = \begin{bmatrix} 0 \\ \boldsymbol{J}^{-1} \end{bmatrix}, \tag{30}$$

where $\boldsymbol{d} = \boldsymbol{\tau}_{\mathrm{act}} + \boldsymbol{\tau}_{\mathrm{p}} + \boldsymbol{\tau}_{\mathrm{d}} + K_q\boldsymbol{e}_q + K_\omega\boldsymbol{\omega}$ is the total deviation from ideal PD torque. It includes allocation and command-saturation errors, not just the neural request. The nonlinear remainder is

$$\boldsymbol{g}(\boldsymbol{x}) = \begin{bmatrix} \frac{1}{2}[(\sqrt{1-\|\boldsymbol{e}_q\|^2}-1)I_3 + [\boldsymbol{e}_q]_\times]\boldsymbol{\omega} \\ -\boldsymbol{J}^{-1}[\boldsymbol{\omega}]_\times\boldsymbol{J}\boldsymbol{\omega} \end{bmatrix} = O(\|\boldsymbol{x}\|^2), \tag{31}$$

with $[\boldsymbol{z}]_\times\boldsymbol{y} = \boldsymbol{z}\times\boldsymbol{y}$. Norms below use fixed coordinates with angular rates expressed in rad/s. The first block's two terms are orthogonal and $(1-\sqrt{1-\|\boldsymbol{e}_q\|^2})^2 \le \|\boldsymbol{e}_q\|^2$, so it is at most $\frac{\sqrt{2}}{2}\|\boldsymbol{e}_q\|\|\boldsymbol{\omega}\|$; since $\boldsymbol{\omega}\times\boldsymbol{J}\boldsymbol{\omega} = \boldsymbol{\omega}\times(\boldsymbol{J}-\mu I_3)\boldsymbol{\omega}$, the choice $\mu = \frac{1}{2}(\lambda_{\max}+\lambda_{\min})(\boldsymbol{J})$ bounds the second by $\frac{1}{2}(\kappa(\boldsymbol{J})-1)\|\boldsymbol{\omega}\|^2$, with κ the condition number. With $\|\boldsymbol{e}_q\|\|\boldsymbol{\omega}\| \le \frac{1}{2}\|\boldsymbol{x}\|^2$,

$$\|\boldsymbol{g}(\boldsymbol{x})\| \le L_r\|\boldsymbol{x}\|^2, \qquad L_r = \tfrac{\sqrt{2}}{4} + \tfrac{1}{2}(\kappa(\boldsymbol{J})-1), \tag{32}$$

i.e., $L_r = 1.048$ (X2) and 0.529 (CF2), above sampled suprema 0.483 and 0.283; the cruder $\sqrt{2}/4+\kappa(\boldsymbol{J})$ also holds.

Theorem 1 (Conditional local practical stability)**.** *Let $\boldsymbol{J}, K_q, K_\omega \succ 0$, let q_d be constant, and use the smooth branch in* (30)*. Choose $Q_L \succ 0$ and the solution $P \succ 0$ of $A^\mathsf{T}P + PA = -Q_L$. On $\|\boldsymbol{x}\| \le r < 1$, suppose $\|\boldsymbol{g}(\boldsymbol{x})\| \le L_r\|\boldsymbol{x}\|^2$ (e.g.,* (32)*) and choose r so that $c = \lambda_{\min}(Q_L) - 2\|P\|L_r r > 0$. Set $p_- = \lambda_{\min}(P)$, $p_+ = \lambda_{\max}(P)$, $b = \|PF\|$, and $V = \boldsymbol{x}^\mathsf{T}P\boldsymbol{x}$. If*

$$V(0) \le p_- r^2, \qquad \|\boldsymbol{d}(t)\| \le \bar{d} < \frac{cr}{2b}\sqrt{\frac{p_-}{p_+}}, \tag{33}$$

then $V \le p_- r^2$ is forward invariant and

$$V(t) \le e^{-\alpha t}V(0) + \frac{4p_+b^2}{c^2}\bar{d}^2, \qquad \alpha = \frac{c}{2p_+}. \tag{34}$$

Thus attitude regulation is locally input-to-state stable; zero perturbation gives local exponential convergence of the plant state.

Proof. Along $\dot{\boldsymbol{x}} = A\boldsymbol{x}$, the mechanical energy $V_L = \boldsymbol{e}_q^\mathsf{T}K_q\boldsymbol{e}_q + \frac{1}{2}\boldsymbol{\omega}^\mathsf{T}\boldsymbol{J}\boldsymbol{\omega}$ satisfies $\dot{V}_L = \boldsymbol{e}_q^\mathsf{T}K_q\boldsymbol{\omega} - \boldsymbol{\omega}^\mathsf{T}K_q\boldsymbol{e}_q - \boldsymbol{\omega}^\mathsf{T}K_\omega\boldsymbol{\omega} = -\boldsymbol{\omega}^\mathsf{T}K_\omega\boldsymbol{\omega} \le 0$ because K_q is symmetric; off-diagonal products of inertia in $\boldsymbol{J}$ do not enter. On the set $\dot{V}_L \equiv 0$, $\boldsymbol{\omega} \equiv \mathbf{0}$ forces $\dot{\boldsymbol{\omega}} \equiv \mathbf{0}$ and hence $\boldsymbol{e}_q = \mathbf{0}$, so by LaSalle's invariance principle [29] A is Hurwitz and the unique $P \succ 0$ exists. Equations (30)–(31) give

$$\dot{V} \le -c\|\boldsymbol{x}\|^2 + 2b\|\boldsymbol{x}\|\bar{d} \le -\frac{c}{2}\|\boldsymbol{x}\|^2 + \frac{2b^2}{c}\bar{d}^2.$$

On $V = p_-r^2$, $\|\boldsymbol{x}\| \ge r\sqrt{p_-/p_+}$, so (33) makes the first bound strictly negative. The ellipsoid stays inside the radius-r ball; integration of $\dot{V} \le -\alpha V + 2b^2\bar{d}^2/c$ gives (34). □

If the same-state baseline deviation from ideal PD is bounded by $\bar{d}_0$, then (29) permits $\bar{d} = \bar{d}_0 + L_A\beta\|\boldsymbol{\ell}\|$ in the theorem, *provided* (33) holds. This is a sufficient local robustness condition, independent of the particular PES trajectory, not a PES convergence proof. Maximizing the admissible bound in (33) over r gives $r^* = \min\{\lambda_{\min}(Q_L)/(4\|P\|L_r), 1\}$ and, when $r^* < 1$,

$$\bar{d}_{\max} = \frac{\lambda_{\min}^2(Q_L)}{16\,b\,\|P\|\,L_r}\sqrt{\frac{p_-}{p_+}}, \qquad \beta < \beta_{\max} = \frac{\bar{d}_{\max} - \bar{d}_0}{L_A\|\boldsymbol{\ell}\|_2}, \tag{35}$$

a sufficient authority bound when $\bar{d}_0 < \bar{d}_{\max}$. With $Q_L = I$ (no $\operatorname{diag}(q_eI_3, I_3)$ did better), A has spectral abscissa -1.395 (X2) and -3.325 (CF2), and the PD command is unsaturated and rotor-feasible in the certified ball, so $\bar{d}_0 = 0$ at hover. Then $(r^*, \bar{d}_{\max}, \beta_{\max}) = (5.31\times10^{-2},\ 3.90\times10^{-4}\,\mathrm{N\,m},\ 9.69\times10^{-5})$ for X2 and $(1.05\times10^{-1},\ 1.11\times10^{-6}\,\mathrm{N\,m},\ 6.43\times10^{-4})$ for CF2 (2.14×10^{-3} with exact allocation). The experimental $\beta = 0.5$ lies far outside this conservative certificate: stability of the reported runs is observed, not certified. Moving references, the 180° sign boundary, sampled-data/solver error, and full translational stability remain outside this result; sample-and-hold error would need its own bound within $\boldsymbol{d}$.

Remark 1 (Sign selection on certified trajectories)**.** *Under Theorem 1, $\|\boldsymbol{e}_q\| \le \|\boldsymbol{x}\| \le r < 1$ gives $|q_{e,0}| \ge \sqrt{1-r^2} >$*

0*: sign selection in* (6) *never switches, the smooth branch is self-consistent, and* $q_{e,0}$ *perturbations below* $\sqrt{1-r^2}$ *cannot flip it; the global obstruction [19] remains.*

a) Nengo realization.: In Nengo 4.1.0 a custom `NeuronType` exposes membrane and gate states, and direct current injection into `ensemble.neurons` bypasses NEF gain/intercept fitting; unfiltered CMG activity uses $\eta_N = \eta(2m)/h$ and Nengo error $\boldsymbol{s}-\boldsymbol{y}^*$, leak uses pre-step weights, and projection follows the simulator step. Nengo's teaching graph adds a one-step latency, so the weight sequences differ; over ten direct/Nengo cases the maximum relative mean-error difference is 2.013×10^{-6}, a tested numerical agreement, not formal backend equivalence.

IV. Experimental Protocol and Cross-Mission Results

A. Models, Parameters, and Measurements

MuJoCo [30] simulates two Menagerie-derived multirotor models [31], CF2 and X2, as free-flying rigid bodies with site-applied wrenches; the attitude loop commands three body torques and a separate height loop commands collective thrust. The upstream CF2 model exposes abstract moment channels [32]; we replace them with four rotor-force channels and the shared clipped inverse allocator (13) to expose thrust/torque coupling and per-rotor limits. Rotor ordering is $(-x,-y),(-x,+y),(+x,+y),(+x,-y)$ with reaction signs $(-,+,-,+)$. The CF2 $1.5\times$ scale (Table I) raises a model limit, not measured hardware capability. Both models are simplified (no motor/RPM/battery states and no validated rotor aerodynamics); feedback uses `qpos`/`qvel`, with noise injected only in Sec. IV-C.

CF2's body axes are principal axes. X2's geometry-derived body tensor, $\boldsymbol{J} = 10^{-2}[3.66517, 0, -.21000;\ 0, 2.54117, 0;\ -.21000, 0, 6.05280]\,\mathrm{kg\,m^2}$ (rows separated by semicolons), differs from both $\mathrm{diag}(\boldsymbol{j}_{\mathrm{tune}})$ and a manufacturer-identified tensor. Torque limits use $\boldsymbol{j}_{\mathrm{tune}}$ as defined in Section II.

Each PD/CMG pair shares initial state, reference, duration, allocator, model, and timestep. The primary campaign is nominal and deterministic (no injected disturbances or sensor noise), one run per cell, learning active throughout. We report the mean error $\bar{\theta}_q$ and the reduction $100(\bar{\theta}_{q,\mathrm{PD}} - \bar{\theta}_{q,\mathrm{CMG}})/\bar{\theta}_{q,\mathrm{PD}}$ over all 90,000 samples, even when plotted traces are decimated.

The five reference triples are generated as follows (radians): Roll is $(0.25\sin 0.7t, 0, 0)$; Full3 is $0.5\sin(t + (0,\pi/2,\pi)^{\mathsf{T}})$, componentwise; Hard45 is $(0.55\sin 0.75t + 0.20\sin(2.10t{+}0.40),\ 0.45\sin(0.95t{+}\pi/3){+}0.18\sin 1.75t,\ 0.65\sin(0.55t + \pi) + 0.25\sin(1.35t + 0.20))$. Hardstep holds zero for one second, then cycles every two seconds through $(.75,-.45,.55)$, $(-.65,.55,-.70)$, $(.50,.50,.90)$, and $(-.80,-.35,.20)$. Yaw360 is $(0, 0, 2\pi(3\xi^2 - 2\xi^3))$ with $\xi = t/45$. Hover is omitted: its near-zero error admits no meaningful percentage.

B. Primary Five-Mission Matrix

CMG-PES improves mean quaternion-angle tracking error in all ten nominal missions while respecting the same actuator interface as the PD baseline. The largest reductions occur on Yaw360 and Full3 and the smallest on Roll. Hardstep has the largest absolute mean error and the only nonzero rotor-bound occupancy S, the logged percentage of samples with at least one rotor command at a bound (PD/CMG): $.016/.239$ on X2 and $.137/.264$ on CF2. The statistic S measures post-clipping commands near a limit, not requested-wrench infeasibility magnitude. The results establish deterministic nominal performance relative to this fixed PD law, not optimality over classical controllers.

Figures 4 and 5 pair X2 Full3 tracking with its internal CMG signals. Improvement refers to the mean metric; it need not hold at every instant.

TABLE I

Model/interface parameters with manufacturer context (top) and canonical 3-channel CMG–PES parameters (bottom). Mass agrees with published CF2/X2 specifications [33], [34]; force limits, inertias, and reaction coefficients are simulation parameters, not identified hardware ratings. Neural states/currents are normalized model variables; learning coefficients are per step.

Parameter	CF2 rotor model	X2 model
Mass M (kg)	0.027	1.325
Rotor $\lvert p_x\rvert, \lvert p_y\rvert$ (m)	$0.0325, 0.0325$	$0.14, 0.18$
Reaction $\lvert\kappa_i\rvert$ (m)	0.005	0.0201
Rotor/total force limit (N)	$.0875 \to .13125$ / $.350 \to .525$	13 / 52
Main-test thrust/weight	1.982	4.001
$\boldsymbol{j}_{\mathrm{tune}}$ ($\mathrm{kg\,m^2}$)	$10^{-5}(2.3951, 2.3951, 3.2347)$	$10^{-2}(6.0711, 3.6468, 2.5412)$
Integrator, timestep	RK4, 0.0005 s	Euler, 0.0005 s
Mfr. thrust data	.568 N[a]	Not published[b]

[a]Historical CF2 measurement at 93.75% PWM using an external power supply [35]; not a continuous battery-powered rating. The modeled .525 N is below that observation. [b]Not reported in the reviewed X2 datasheet. Neither model includes motor lag; a historical CF2 motor step took about 180 ms [36], illustrating an omitted hardware effect.

CMG–PES	Value		Value
h, duration	0.0005 s, 45 s	$\alpha_h, \gamma, g_0, g_1$	$50, 2, 120, 30$
τ, τ_r	0.05 s, 0.005 s	I_b, I_g, G	$0.75, 1.25, 10^5$
$v_{\mathrm{rest}}, v_{\mathrm{reset}}, v_{\mathrm{th}}$	$0, 0, 1$	β, η	$0.5, 8\times10^{-4}$
$v_{j,0}, r_{j,0}, \chi$	$0, 0, 1$	$\lambda_W, W_{\max}, k_e, k_\omega^*$	$10^{-6}, 6, 8, 0$

C. Sensitivity to Measurement Noise

Each of the ten cases was rerun at the two noise levels of Table II with two paired seeds: at every step, independent zero-mean Gaussian samples form a small Euler-angle quaternion left-multiplied into q and corrupt each angular-rate component. These levels are not calibrated sensor specifications, and the runner evaluates θ_q from the *corrupted* quaternion used by the controller, not a ground-truth attitude, so the Mod./Real. columns measure noisy-feedback error. All noisy reductions remain positive, spanning 18.08–55.75% at the higher level; Yaw360 loses about 12.8 pp relative to clean feedback, whereas Roll's reduction increases. Seed SDs are .001–.101 pp (moderate) and .002–.031 pp (realistic): limited repeatability, not a population confidence interval.

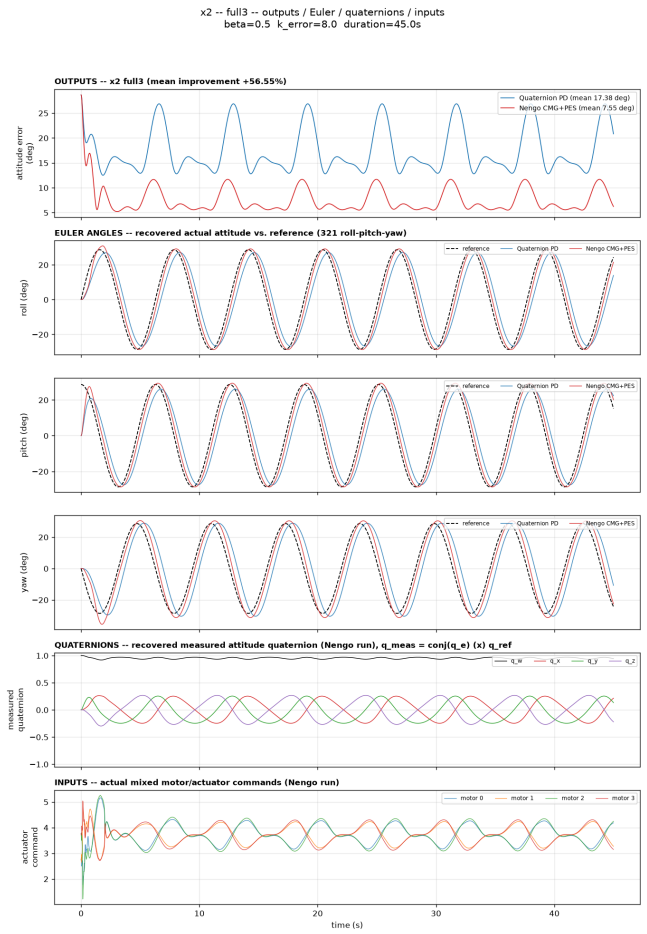


Fig. 4. X2 Full3 mission: PD baseline versus CMG–PES augmentation over the 45-s run. Full3 commands all three axes simultaneously and is the largest reduction that is also insensitive to measurement noise, losing under one point between the clean and realistic columns of Table II.

D. Ablation: What Drives the Improvement?

We separate decoder adaptation (PES), input-dependent conductance modulation (χ), and readout gain (G). Toggling χ, which retains the base shunt $g_0 r_j^2$, is numerically indistinguishable with the decoder mode held fixed. *Fixed versus adaptive decoder.* Holding $W_k = W_0$ keeps the neuron dynamics but reduces the readout to $\boldsymbol{a}^+ - \boldsymbol{a}^-$, a bounded, stateful image of the PD request. It still helps (X2 0.12–10.27%, CF2 0.06–6.80%; Table II), so fixing the decoder does not remove *all* improvement, yet it recovers at most a fifth of the PES gain in any cell. Fig. 5 shows why, row by row. *Membrane and gate.* At $I_j = I_b$ the membrane rests at $v_j \approx 0.75 < v_{\mathrm{th}}$, so $r_j = \sigma(\alpha_h(I_b - v_{\mathrm{th}})) = \sigma(-12.5) \approx 3.7 \times 10^{-6}$: the gate is deep in its logistic tail, not closed. Once $r_j \geq 1/2$, $\gamma = 2$ and $v_j \geq 0$ (Proposition 1) give $b_{j,k} \geq 1/2$ in (20), so an opened gate cannot close (none did in the twenty runs), and the shunt holds v_j near 0.09. *Activity.* With $G = 10^5$ the resting gate already gives $a_j \approx 0.37$, graded by the input; a latched gate pins $a_j = 1$. With the fixed decoder only the $\pm y$ pair latches on Full3 (both models), so $a_y^+ - a_y^- = 0$ and the pitch residual is

TABLE II

CANONICAL NENGO CMG–PES VERSUS PD: MEAN QUATERNION-ANGLE ERROR [RAD] AND REDUCTION VS PD [%], 45 S, $n = 1$ PER CLEAN CELL. PES: CMG–PES; FIXED: SAME UNITS WITH $W_k = W_0$; MOD./REAL.: CMG–PES UNDER THE MODERATE (0.010 RAD, 0.025 RAD S^{-1}) AND REALISTIC (0.020 RAD, 0.050 RAD S^{-1}) ATTITUDE/RATE NOISE OF SEC. IV-C, TWO-SEED MEANS WITH SEED SD ≤ 0.03 PP AT THE REALISTIC LEVEL. CF2 DENOTES THE 1.5× ROTOR-AUTHORITY DESIGN, NOT STOCK HARDWARE.

		Error [rad]		Reduction vs PD [%]			
Model	Mission	PD	CMG–PES	PES	Fixed	Mod.	Real.
X2	Roll	.050565	.041785	17.36	1.93	32.3	32.2
	Full3	.303275	.131748	56.56	9.74	56.4	55.7
	Hard45	.294082	.138007	53.07	10.27	52.9	52.3
	Hardstep	.454528	.359355	20.94	0.46	20.8	20.6
	Yaw360	.075863	.029902	60.58	0.12	56.8	47.8
CF2	Roll	.051337	.042439	17.33	3.31	32.4	32.5
	Full3	.288383	.126476	56.14	4.53	55.9	55.2
	Hard45	.285449	.132319	53.65	6.80	53.4	52.7
	Hardstep	.426182	.347886	18.37	0.42	18.3	18.1
	Yaw360	.075812	.029870	60.60	0.06	56.8	47.8

exactly zero after 1 s while the sub-threshold units carry roll and yaw. Under PES on X2, $\boldsymbol{a}_k \equiv \mathbf{1}$ from 2.02 s (all gates latched by 2.4 s), and for constant $\bar{\boldsymbol{a}}$ with inactive projection the direct update gives $\boldsymbol{z}_k = W_k \bar{\boldsymbol{a}}$ the recursion

$$\boldsymbol{z}_{k+1} = (1 - \lambda_W)\boldsymbol{z}_k + \eta \|\bar{\boldsymbol{a}}\|^2 [\boldsymbol{y}_k^* - \mathrm{clip}_{[-1,1]}(\boldsymbol{z}_k)], \quad (36)$$

the leaky constant-feature regime of Proposition 3: a first-order filter of the teaching target with pole 0.9952 (about 0.104 s) while the readout clip is inactive. With $\|\bar{\boldsymbol{a}}\|^2 = 2m$ its DC gain is $6\eta/(6\eta + \lambda_W) \approx 0.9998$: the latched residual is *lagged proportional* action, not integral action (which needs a unit pole). For $|k_e e_{q,i}| < 1$ it adds $\beta k_e \ell_i / k_{q,i} = \beta k_e \nu_i / (2\omega_{n,i}^2)$, i.e., 1.50× (roll, pitch) and 1.63× (yaw) the PD stiffness on both vehicles, behind a 9.60 rad/s corner. The smooth references are sums of constant-frequency components [37]; their largest instantaneous frequency, 2.1 rad/s (Hard45), lies below the corner (gain 0.977, lag 12.3°), while noise in $\boldsymbol{e}_q$ above it is attenuated. *Residual and decoder.* The residual RMS rises from 0.026 to 0.123 N m and the decoder moves $\|W_T - W_0\|_F = 0.37$; on Roll, Full3, Hard45, and Hardstep PES applies 4–12× the fixed residual RMS on both models. Where no gate opens (Roll, Yaw360; $r_j \leq 1.6 \times 10^{-3}$) PES lifts the driven membranes toward threshold ($v_j \leq 0.87$) and uses the graded tail. Decoder adaptation is therefore the principal lever: the CMG dynamics supply a bounded basis, latched or graded, on which PES shapes the correction. The leak λ_W is the σ-modification of robust adaptive control [38], [39]: at a fixed point of (25) with constant $\bar{\boldsymbol{a}}$ and inactive clip and projection, $\eta \boldsymbol{\varepsilon} \|\bar{\boldsymbol{a}}\|^2 = \lambda_W \boldsymbol{s}$, an exact PES mismatch of $2.1 \times 10^{-4} \|\boldsymbol{s}\|$ latched and $1.5 \times 10^{-3} \|\boldsymbol{s}\|$ at the resting tail.

E. Neuron-Family Comparison

The Nengo benchmark covers two models, five missions, and eleven controllers: PD and fixed/PES variants of CMG,

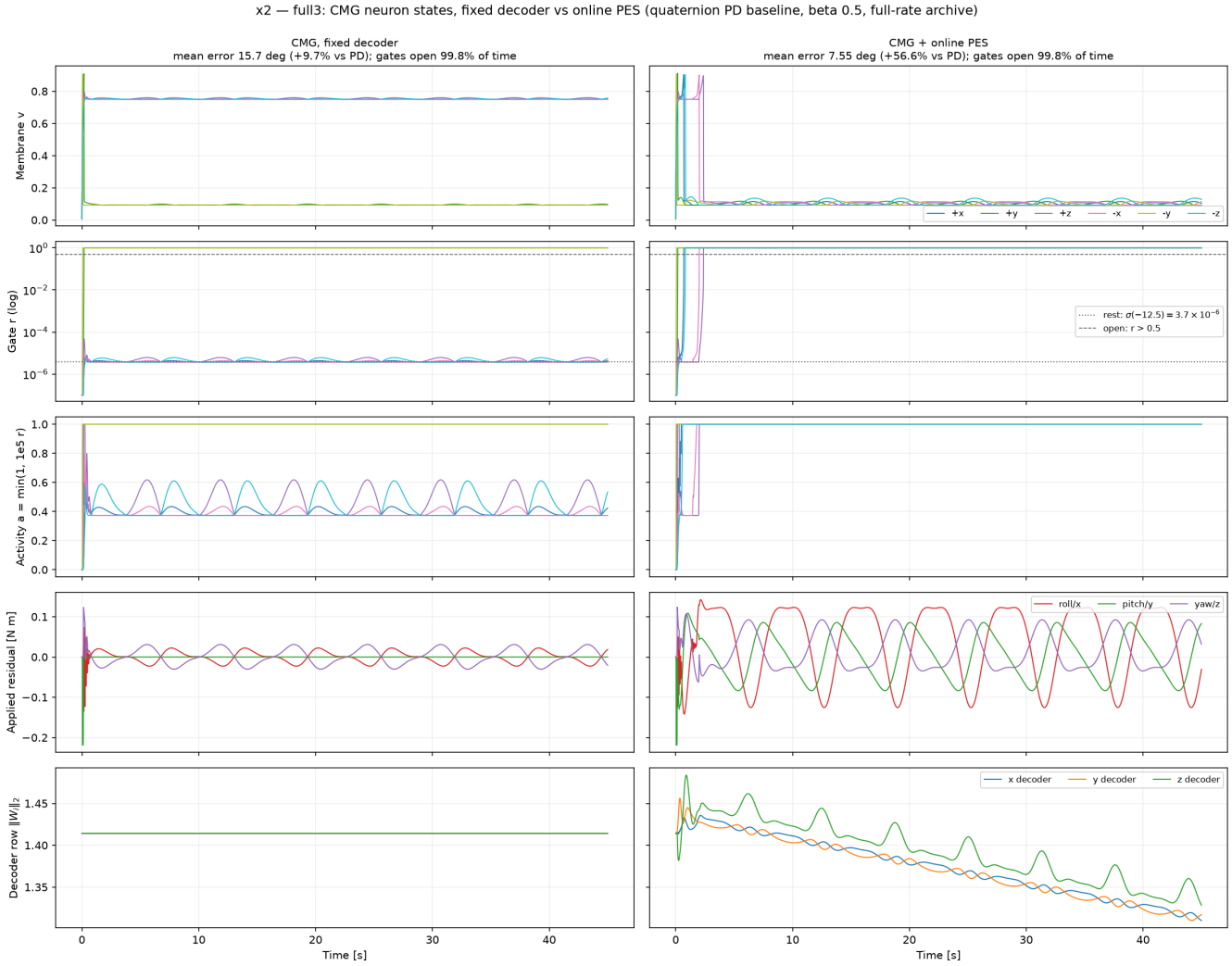


Fig. 5. X2 Full3: internal CMG signals on the trajectory of Fig. 4, comparing fixed-decoder CMG (left) to online PES adaptation (right). Rows separate membrane state, raw gate r_j, and the clipped activity a_j presented to the decoder.

LIF, ALIF, Izhikevich [40], and LIFRate, all sharing the bounded residual, saturation, and allocation interface; only the neuron dynamics and native-state-to-activity map differ. Fig. 6 shows two patterns. First, PES is the dominant lever in every family: fixed decoders rarely match their PES counterparts, even with more expressive dynamics, extending Sec. IV-D to every basis. Second, lower error is not free: some Izhikevich–PES cells reach the smallest error, but at an average normalized residual RMS near 0.37 versus 0.12 for CMG–PES and with larger decoder movement, so equal output caps do not imply equal realized effort. Proposition 3 explains the flat rows: its factor $1 - \eta\|\boldsymbol{a}\|^2$ stays near one on a silent basis. LIF, ALIF, and LIFRate units are active ($|a_j| > 10^{-9}$) in 0.1% of unit-samples on Full3 and Hard45, under 0.05% on Roll and Yaw360, and 6–20% on Hardstep, so their gains stay below 0.1% outside Hardstep; CMG units (resting near $a_j = 0.37$ on the logistic tail) and Izhikevich units are active in $\geq 99.9\%$, and both learn. Persistent activity is thus necessary for PES here but does not set effort, since Izhikevich–PES shares it. CMG–PES attains comparable tracking at markedly lower realized effort and more conservative decoder motion.

F. Disturbance Rejection and Retention

An equilibrium test applies $\boldsymbol{\tau}_{\mathrm{d}} = d_b \rho(t)[\sin(1.31t + .2), .8\sin(1.73t + 1.1), .6\sin(2.17t + 2)]^{\mathsf{T}}$ in body axes. After 5 s of quiet, ρ alternates 5-s on/off intervals with 0.25-s raised-cosine edge ramps. For X2, $d_b = .15$ N m is the first component's amplitude, not a vector-norm bound. The four-burst 45-s experiment (Fig. 7) cuts burst-phase mean error from $11.50°$ to $6.65°$ (42.2%) and post-burst quiet error from $1.21°$ to $.99°$ (18.0%). The CF2 rerun uses $d_b = .00015$ N m (a selected scaling, not an exact inertia-ratio match); its full-run mean falls from .141199 to .070259 rad (50.24%).

V. Discussion and Conclusion

A bounded residual of six continuous CMG units and eighteen PES-adapted decoder weights augments a fixed quaternion PD baseline without changing its gains or actuator



Fig. 6. Mechanism separation across the Nengo neuron-family benchmark, X2 (top) and CF2 (bottom). Left: mean q-error improvement over the PD baseline. Right: normalized residual effort, the RMS residual as a fraction of the per-axis limit. Rows are neuron family and learning mode, columns are missions; every cell uses the same bounded residual, saturation, and allocation interface. Two patterns hold on both platforms: PES adaptation, not the neuron model, separates the improving rows from the flat ones; and the families that match or exceed CMG–PES on error do so at three to twenty times its residual effort.

interface, reducing mean quaternion-angle error by 17.3–60.6% over ten matched 45-s MuJoCo cases on two modeled platforms, within rotor saturation and with exact baseline recovery at zero authority; direct and Nengo realizations agree to 2.013×10^{-6} relative mean error.

Decoder adaptation is the dominant source of improvement in the tested regime: a fixed decoder gives at most 10.3%, and toggling χ is indistinguishable. Opened gates latch and saturate the readout, where a fixed paired decoder can cancel an axis and PES reduces to lagged proportional action worth about 1.5–1.6× the PD stiffness, while subthreshold units stay in a graded logistic tail; PES shapes the correction on either basis as long as the basis stays active. Across all five neuron families PES is the consistent differentiator, and the dynamics mainly set an effort–accuracy tradeoff. The readout gain G, which sets the resting activity at 0.37, is a key sensitivity parameter. The local certificate gives $\beta_{\max} \sim 10^{-4}$–$10^{-3}$, far below the tested $\beta = 0.5$.

Limitations: results are in simulation (Sec. IV-C bounds noise sensitivity; model mismatch and hardware effects are not addressed), the retention horizon is limited, and bounded commands, states, and weights are implementation properties, not global stability proofs.

Future work includes baselines with desired-rate feedforward, integral action, and matched stiffness, matched-effort comparisons against non-neural bounded bases, sweeps of G and χ, less conservative sampled-data certificates, longer retention studies, held-out evaluations with frozen learned weights, full 6-DOF extension, and, most importantly, validation on physical aerial, ground, and other robotic platforms; the same bounded-residual wrapper extends to nonlinear and optimal-control baselines with hard actuator constraints.

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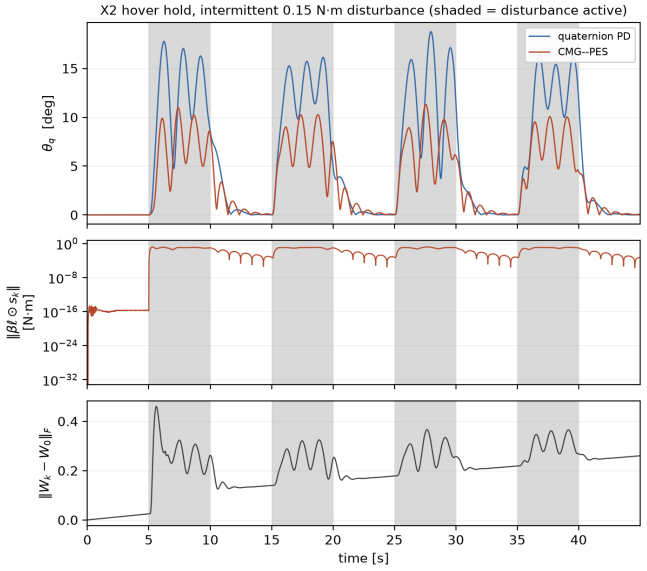


Fig. 7. X2 hover hold under four $0.15\,\mathrm{N\,m}$ disturbance bursts (shaded) separated by quiet intervals. Top: attitude error. Centre: residual magnitude, logarithmic. Bottom: decoder displacement from W_0. The residual is at numerical precision before the first burst. This means negligible initial correction, not zero computation or energy cost.

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